

PROJECT TITLE 1

SMART REACTION CARNIVAL WHACK A LIGHT

NAMES OF GROUP MEMBERS

1. _____
2. _____
3. _____
4. _____
5. _____

BRIEF DESCRIPTION OF THE PROJECT

Smart Reaction Carnival Whack a Light is an IoT-based target game in which a player throws small, lightweight balls at illuminated targets. Several impact targets are arranged on a vertical board. The ESP32 randomly activates one target at a time, and the player must hit the illuminated target before the time limit expires. A successful hit adds points, while a missed target or an expired target records a miss. The game automatically displays the remaining time and final score.

HOW THE PROJECT INCORPORATES THE CARNIVAL THEME

The project adapts the familiar carnival target booth into an electronic reaction game. Bright target lights, sound effects, a countdown, printed carnival graphics, and lightweight throwing balls create a game that visitors can understand immediately. The target board can be decorated with circus colors, ticket patterns, stars, and point values while the electronics provide automatic scoring and feedback.

PROJECT OBJECTIVES

- Develop a safe ball-throwing carnival game controlled by an ESP32 Development Kit.
- Detect valid hits on randomly illuminated targets and calculate the score automatically.
- Measure response time, correct hits, misses, and the player's final score.
- Provide visual and audible feedback through target lights, a display, and a buzzer.
- Prepare the game to transmit session data to intelligent-system software in a later development stage.

IOT COMPONENTS AND TECHNOLOGIES TO BE USED

The ESP32 serves as the main microcontroller and controls the random target sequence, countdown timer, hit validation, and score computation. Each target uses a large impact plate connected to a microswitch, limit switch, or protected arcade button. LEDs identify the active target, while an OLED display and buzzer communicate the game status. The ESP32's built-in Wi-Fi capability can later send game records to the intelligent-system software.

EXPECTED FUNCTIONALITY

- The operator presses the start button to begin a new game.
- The display presents a short countdown before the first target activates.
- The ESP32 randomly illuminates one target at a time.
- The player throws a lightweight foam ball at the illuminated target.
- A hit on the correct target within the allowed time adds points.
- A wrong hit or an expired target records a miss and does not add points.
- The game continues until the round timer reaches zero.
- The display shows the final score, correct hits, misses, and average response time.
- The reset control prepares the system for the next player without restarting the ESP32.

FUTURE INTELLIGENT SYSTEM INTEGRATION

- Player Profile and Leaderboard - The software will store player names, scores, and session history.
- Adaptive Difficulty - The software will use reaction time and accuracy to recommend or select a suitable target speed.
- Performance Analysis - The software will summarize correct hits, misses, average response time, and improvement across sessions.

DATA TO BE COLLECTED

- Player or session identifier
- Final score
- Correct target hits
- Wrong target hits
- Expired targets or misses
- Response time for each target
- Average and fastest response time

- Difficulty level and round duration

INITIAL LIST OF HARDWARE AND COMPONENTS NEEDED

- 1 ESP32 Development Kit
- 1 breadboard for initial testing
- 1 perforated PCB or solderable prototype PCB for the final circuit
- 4 to 6 large impact plates with microswitches, limit switches, or protected arcade buttons
- 4 to 6 LED modules, illuminated button rings, or addressable RGB LED sections
- 1 I2C OLED display
- 1 active or passive buzzer
- 1 start button and 1 reset button
- Resistors, capacitors, pin headers, screw terminals, and connectors
- Jumper wires and insulated hook-up wires
- 1 regulated 5 V power supply or power bank
- Lightweight foam balls
- Plywood for the target face and structural supports
- Cardboard, foam board, or illustration board for the carnival enclosure
- Optional clear acrylic sheets for target protection
- Hot glue, screws, nuts, cable ties, electrical tape, and heat-shrink tubing
- Paint, colored paper, printed labels, and carnival decorations

ESTIMATED BUDGET

The estimated cost for an exhibition-ready build is PHP 2,200 to PHP 3,200. A working target with fewer lights and recycled construction materials may cost less. A group budget of approximately PHP 3,000 is recommended to cover the final structure and replacement components.

ONE MONTH DEVELOPMENT SCHEDULE

- Week 1 - Test one target, finalize the scoring rules, and purchase the required components.
- Week 2 - Complete the target-control program, timing system, display, and score calculation.
- Week 3 - Transfer the circuit to a perforated PCB and construct the target board and enclosure.
- Week 4 - Test repeated rounds, adjust target sensitivity, add decorations, and practice the demonstration.

SAFETY AND RELIABILITY CONSIDERATIONS

Only lightweight foam balls will be used. The target board will have a stable base, covered wiring, rounded edges, and protected electronic components. The project will operate from a regulated low-voltage power supply. Ordinary exposed push buttons will not receive direct ball impacts; each electronic switch will be protected by a larger impact plate.

PROJECT TITLE 2

SMART BEANBAG TOSS

NAMES OF GROUP MEMBERS

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2. _____
3. _____
4. _____
5. _____

BRIEF DESCRIPTION OF THE PROJECT

Smart Beanbag Toss is an IoT-based carnival game that automatically detects and scores beanbags thrown through different target holes. Each scoring hole contains an infrared sensor positioned inside its return channel. When a beanbag passes through a hole, the ESP32 identifies the scoring zone, adds the assigned points, and updates the display. The player receives a fixed number of beanbags or plays within a set time limit.

HOW THE PROJECT INCORPORATES THE CARNIVAL THEME

The project uses the appearance and rules of a traditional carnival throwing booth. The target board can include colorful rings, different point values, character illustrations, lights, and sound effects. The physical throwing activity attracts participants, while automatic scoring gives the booth a clear IoT function.

PROJECT OBJECTIVES

- Construct a carnival beanbag game with electronic target detection and automatic scoring.
- Use infrared sensors to identify the scoring hole entered by each beanbag.
- Record the player's score, successful throws, missed throws, and scoring-zone accuracy.
- Provide immediate feedback using LEDs, a buzzer, and an OLED display.
- Prepare the system to share game data with intelligent-system software at a later stage.

IOT COMPONENTS AND TECHNOLOGIES TO BE USED

The ESP32 reads the infrared sensors installed inside the scoring channels and processes each valid throw. It applies a short lockout period after every detection to prevent one beanbag from being counted more than once. LEDs show the active game

state, the buzzer confirms successful throws, and the OLED displays the score, remaining throws, and timer. Wi-Fi communication can later transfer session results to external software.

EXPECTED FUNCTIONALITY

- The operator starts a new game and selects a timed round or a fixed number of throws.
- The display shows the starting score, remaining time, or remaining beanbags.
- The player throws a beanbag toward the scoring holes.
- An infrared sensor detects the beanbag inside the selected scoring channel.
- The ESP32 adds the point value assigned to that hole and activates feedback lights and sound.
- The software lockout prevents duplicate scoring while the same beanbag moves through the channel.
- The game ends when the timer expires or the allowed throws are completed.
- The display presents the final score, successful throws, misses, and most frequently reached scoring zone.
- A reset control clears the previous session for the next player.

FUTURE INTELLIGENT SYSTEM INTEGRATION

- Player Profile and Leaderboard - The software will store player scores, rankings, and previous game sessions.
- Accuracy Analysis - The software will calculate success rates and identify the scoring zones reached most often.
- Adaptive Challenge - The software will recommend different time limits, throwing distances, or target values based on player performance.

DATA TO BE COLLECTED

- Player or session identifier
- Final score
- Number of successful throws
- Number of missed throws
- Detected scoring zone for each throw
- Points earned per throw
- Round duration or number of allowed throws
- Scoring-zone accuracy

INITIAL LIST OF HARDWARE AND COMPONENTS NEEDED

- 1 ESP32 Development Kit
- 1 breadboard for initial testing
- 1 perforated PCB or solderable prototype PCB for the final circuit
- 3 to 5 infrared obstacle or break-beam sensors
- 3 to 5 target LEDs or addressable RGB LED sections
- 1 I2C OLED display
- 1 active or passive buzzer
- 1 start button and 1 reset button
- Resistors, capacitors, pin headers, screw terminals, and connectors
- Jumper wires and insulated hook-up wires
- 1 regulated 5 V power supply or power bank
- 3 to 6 lightweight beanbags
- Plywood or another rigid material for the target face
- Cardboard or PVC channels for returning and separating beanbags
- Cardboard, foam board, or illustration board for the collection box and carnival enclosure
- Optional clear acrylic for protecting sensors and displays
- Hot glue, screws, nuts, cable ties, electrical tape, and heat-shrink tubing
- Paint, colored paper, printed point labels, and carnival decorations

ESTIMATED BUDGET

The estimated cost for an exhibition-ready build is PHP 2,500 to PHP 3,700. The final amount depends on the number and type of infrared sensors and whether plywood must be purchased. A group budget of approximately PHP 3,500 is recommended for a stable target structure and spare sensor components.

ONE MONTH DEVELOPMENT SCHEDULE

- Week 1 - Test one infrared scoring channel, finalize the target dimensions, and purchase components.
- Week 2 - Complete the scoring program, duplicate-detection prevention, display, timer, and feedback system.
- Week 3 - Transfer the circuit to a perforated PCB and build the target face, scoring channels, and collection box.
- Week 4 - Test different beanbag sizes and throwing angles, secure the sensors, decorate the booth, and rehearse the demonstration.

SAFETY AND RELIABILITY CONSIDERATIONS

The game will use soft beanbags and a clearly marked throwing line. The target structure will be secured against tipping, and the area behind it will remain clear. Wires and sensors will be placed inside protected channels. The system will operate from a regulated low-voltage supply, and all rigid edges will be covered or rounded.

PROJECT TITLE 3

SPEED TAP CHALLENGE

NAMES OF GROUP MEMBERS

1. _____
2. _____
3. _____
4. _____
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BRIEF DESCRIPTION OF THE PROJECT

Speed Tap Challenge is an IoT-based carnival game that measures how many valid taps a player can complete during a short timed round. The player repeatedly presses a large arcade button or taps a protected impact pad. The ESP32 counts valid inputs, measures the interval between taps, and displays the final tapping speed. Indicator lights and sound effects provide immediate feedback during the game.

HOW THE PROJECT INCORPORATES THE CARNIVAL THEME

The project presents a fast-action carnival challenge in which players compete for the highest tap count. A colorful strength-meter-style display, countdown lights, sound effects, printed high-score labels, and a large tapping target create the appearance of a compact carnival booth. The short round duration encourages visitors to replay the game and compare scores.

PROJECT OBJECTIVES

- Develop a timed tapping game controlled by an ESP32 Development Kit.
- Count valid taps accurately while preventing switch bounce and false inputs.
- Measure total taps, tapping rate, timing consistency, and changes in speed during the round.
- Display game information and provide visual and audible feedback.
- Prepare the collected data for later intelligent-system analysis.

IOT COMPONENTS AND TECHNOLOGIES TO BE USED

The ESP32 reads a large arcade button or a protected piezoelectric impact sensor and validates every tap. It handles switch debouncing, maintains the countdown, calculates taps per second, and updates an OLED display and LED meter. A buzzer signals the beginning and end of the round. The ESP32 can later send session records through Wi-Fi to the intelligent-system software.

EXPECTED FUNCTIONALITY

- The operator or player presses the start button to begin a new game.
- The display presents a short countdown before input is accepted.
- The player presses the arcade button or impact pad as rapidly as possible.
- The ESP32 validates each tap and rejects signals caused by switch bounce.
- The display and LED meter update while the timer is running.
- The buzzer signals when the round has ended.
- The display shows the total taps, taps per second, fastest interval, and consistency result.
- The reset control prepares the game for the next player.

FUTURE INTELLIGENT SYSTEM INTEGRATION

- Player Profile and Leaderboard - The software will store player scores and rank tapping performance.
- Performance Classification - The software will classify tapping speed and consistency using collected session data.
- Fatigue and Improvement Analysis - The software will compare early and late tapping rates and provide a performance summary.

DATA TO BE COLLECTED

- Player or session identifier
- Total valid taps
- Round duration
- Taps per second
- Time interval between taps
- Fastest and slowest tap interval
- Tapping rate during the beginning and end of the round
- Invalid or rejected inputs

INITIAL LIST OF HARDWARE AND COMPONENTS NEEDED

- 1 ESP32 Development Kit
- 1 breadboard for initial testing
- 1 perforated PCB or solderable prototype PCB for the final circuit
- 1 large arcade button or 1 protected piezoelectric impact sensor
- 1 start button and 1 reset button
- 1 I2C OLED display
- 1 LED bar graph or addressable RGB LED strip

- 1 active or passive buzzer
- Resistors, capacitors, pin headers, screw terminals, and connectors
- Input-protection components if a piezoelectric sensor is used
- Jumper wires and insulated hook-up wires
- 1 regulated 5 V power supply or power bank
- Plywood for the tapping surface and internal support
- Cardboard, foam board, or illustration board for the carnival enclosure
- Optional clear acrylic for the LED meter and display cover
- Foam or rubber padding around the tapping area
- Hot glue, screws, nuts, cable ties, electrical tape, and heat-shrink tubing
- Paint, colored paper, printed labels, and carnival decorations

ESTIMATED BUDGET

The estimated cost for an exhibition-ready build is PHP 1,600 to PHP 2,400. Using a large arcade button instead of a calibrated impact sensor can reduce development risk. A group budget of approximately PHP 2,200 is recommended to cover the final enclosure and replacement parts.

ONE MONTH DEVELOPMENT SCHEDULE

- Week 1 - Test the tapping input, select the round duration, and purchase the required components.
- Week 2 - Complete the tap-validation program, timer, score calculations, LED meter, and display.
- Week 3 - Transfer the circuit to a perforated PCB and construct the tapping panel and carnival enclosure.
- Week 4 - Test with different players, adjust input sensitivity or debounce timing, decorate the game, and rehearse the demonstration.

SAFETY AND RELIABILITY CONSIDERATIONS

The tapping surface will be securely mounted and padded around its edges. Players will use an open hand or finger rather than a hard object. The button or sensor will be protected from excessive force, and the enclosure will prevent access to wiring. The system will operate from a regulated low-voltage power supply.